Functional Autoencoders with Temporal Latent Dynamics for Forecasting

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I. Background

- Functional data arise in many applications such as fertility rates, biomedical signals, and environmental measurements.
- Classical approaches such as Functional Principal Component Analysis (FPCA) provide linear dimension reduction. However, FPCA cannot capture nonlinear structure.
- Goal: Develop a framework that: learns **nonlinear representations**, preserves **smoothness**, and supports **accurate forecasting**

II. Methodology: Functional Autoencoder

We propose a two-stage framework that integrates representation learning with forecasting. (Hsieh, Sun, Wang, & Honavar, 2021), (Wu, Beaulac, & Cao, 2024)

i. Step 1: Representation Learning

Let $x_t = (X_t(s_1), ..., X_t(s_J))^\top$

- Encoder: $h_t = E_\theta(x_t)$
- Decoder (basis expansion): $\hat{X}_t(s) = \sum_{m=1}^{M} \beta_{tm}\phi_m(s)$
- Reconstruction: $\hat{x}_t = \Phi C h_t$

ii. Penalized Objective

$$\mathcal{L} = \frac{1}{T}\sum_{t=1}^{T} \parallel x_t - \hat{x}_t \parallel^2 + \lambda \int \left(\frac{d^2}{ds^2}\hat{X}_t(s)\right)^2 ds \qquad (1)$$

- Ensures: smoothness, noise reduction, and stable representations

iii. Step 2: Latent Temporal Modeling

Latent series: $\{h_t\}_{t=1}^{T}$
Modeled via VAR: $h_t = \sum_{i=1}^{p} A_i h_{t-i} + \varepsilon_t$
Forecast: $\hat{h}_{T+1}, ..., \hat{h}_{T+L}$

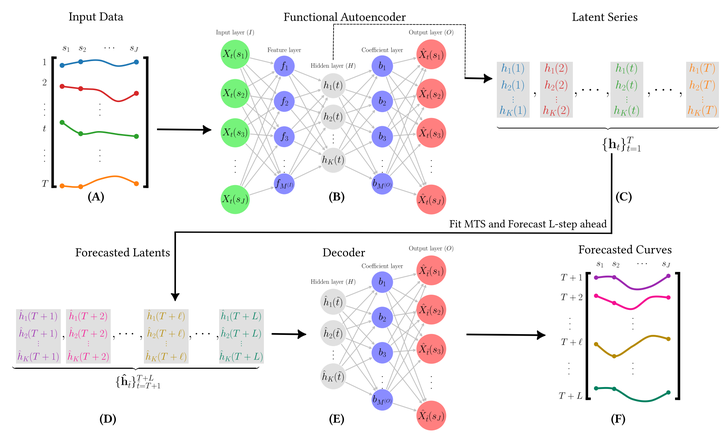
iv. Step 3: Decode Forecasts $\hat{X}_{T+\ell}(s) = D_\theta\left(\hat{h}_{T+\ell}\right)$

III. Algorithm

Functional Autoencoders with Temporal Latent Dynamics

1. **Input:** Functional data $\left\{X_{t(.)}\right\}_{\{t=1\}}^{T}$
2. **Train Autoencoder**
   - Encode: $h_t = E_{\theta(x_t)}$
   - Decode: $\hat{x}_t = D_{\theta(h_t)}$
   - Minimize penalized reconstruction loss
3. **Latent Modeling:** Fit VAR model on $\{h_t\}$
4. **Forecast** Compute $\hat{h}_{T+1}, ..., \hat{h}_{T+L}$
5. **Decode Forecasts:** $\hat{X}_{T+\ell}(s) = D_{\theta(\hat{h}_{T+\ell})}$

IV. Proposed Framework

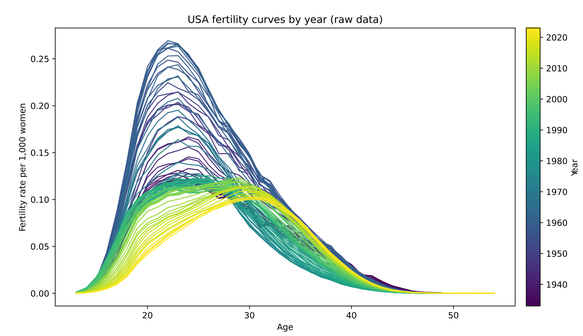


V. Simulation Study

Comparison Setup: FPCA + VAR (linear baseline) and FAE + VAR (proposed)
Scenarios
• Linear setting ⟶ both methods perform similarly • Nonlinear setting ⟶ includes warping, deformation, nonlinear dynamics
Key Finding: FAE captures nonlinear structure and FPCA struggles with phase variation

VI. Real Data Application

USA Fertility Data: Age-specific fertility rates in the U.S. (1933–2023) are observed as smooth age curves with strong temporal dependence, making them well-suited for functional time series modeling.(Hyndman & Ullah, 2007)



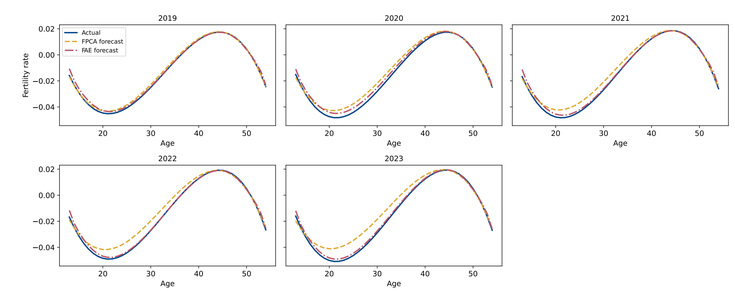
VII. Results

i. Forecasting Setup

Data Split: Training period: 1933–2018. Testing period: 2019–2023
Forecasting Frameworks
• FPCA ⟶ VAR ⟶ Reconstruction (linear baseline) • FAE ⟶ VAR ⟶ Decode (proposed nonlinear approach)
Objective: Compare linear and nonlinear representations.

ii. Forecasting Results

FPCA+VAR: relRMSE = 0.195 \ FAE+VAR: relRMSE = 0.076



Key Observations
• FAE significantly reduces forecasting error
• Better captures peak fertility age
• Preserves full curve shape
• Tracks temporal evolution more accurately than FPCA
• Demonstrates advantage of nonlinear latent representation

VIII. Conclusion

- FAE provides nonlinear functional representations
- Significantly improves forecasting accuracy
- Combines: Functional Data Analysis, Deep Learning, and Time-Series Modeling
- Applicable to: Biomedical data, Environmental data, and Demographic studies

IX. Reference

Hsieh, T. Y., Sun, Y., Wang, S., & Honavar, V. (2021). Functional autoencoders for functional data representation learning. *SIAM international conference on data mining* (pp. 666–674).

Hyndman, R. J., & Ullah, M. S. (2007). Robust forecasting of mortality and fertility rates: A functional data approach. *Computational Statistics & Data Analysis, 51*(10), 4942–4956.

Wu, S., Beaulac, C., & Cao, J. (2024). Functional autoencoder for smoothing and representation learning. *arXiv preprint arXiv:2401.09499*. Retrieved from https://arxiv.org/abs/2401.09499